Addenda (dry): Unitary core, ABS-first readout, prime-log spectrum, and explicit tail control

0. Scope and non-claim (important)

We use standard functional-analytic language to define a coherent operator pipeline and explicit analytic bounds. No claim is made that any construction produces physical "free energy" or violates conservation laws. When we use phrases like "residue" or "seed", they are *structural* witnesses in a readout channel (correlation/information), not a perpetual-motion source.

1. Hilbert-space pipeline: sectors, phases, flow, scaling, measurement

Let $\mathcal{H}$ be a complex Hilbert space. We formalize the "four sectors" as orthogonal projections $P_1, \dots, P_4 : \mathcal{H} \to \mathcal{H}$ such that

$$P_i^2 = P_i, \quad P_i^* = P_i, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_{i=1}^{4} P_i = I. \tag{1}$$

Let $\{\psi_{i,k,m,n}\} \subset \mathcal{H}$ be elementary modes and $c_{i,k,m,n} \in \mathbb{C}$ coefficients. Define a baseline synthesis

$$\Psi_0 \ = \sum_{i=1}^{4} \sum_{k,m,n} c_{i,k,m,n} \, P_i \psi_{i,k,m,n}. \tag{2}$$

A "wave on top" is represented by harmonic phase weights

$$\Psi_{\mathrm{osc}}(t) \ = \sum_{i=1}^{4} \sum_{k,m,n} c_{i,k,m,n} \, e^{i\omega_n t} \, P_i \psi_{i,k,m,n}. \tag{3}$$

The "everything shifting" component is a unitary flow $U(\Delta(t))$:

$$U(\Delta(t))^* U(\Delta(t)) = I, \qquad \text{typically } U(\Delta(t)) = e^{-i\Delta(t)G}, \ G = G^*. \tag{4}$$

A standard scaling action is a unitary dilation on $L^2(\mathbb{R})$:

$$(D_\lambda f)(x) = \lambda^{1/2} f(\lambda x), \qquad \lambda > 0, \tag{5}$$

with dyadic scale $\lambda = 2^j$ and log-parameter $s = \log \lambda$. The contraction step "by factor 2" corresponds to

$$\lambda = \tfrac{1}{2}, \qquad s = \log(1/2) = -\log 2. \tag{6}$$

Define the full internal state (pipeline)

$$S(t) := \Psi_{\mathrm{osc}}(t), \qquad \Psi(t) := U(\Delta(t)) \, D_{\lambda(t)} \, S(t). \tag{7}$$

ABS-first readout. The measured output is a scalar, taken through an absolute value (branch-safe near nodes):

$$A_{\mathrm{norm}}(t) := \|\Psi(t)\|, \qquad A_\phi(t) := |\langle \phi, \Psi(t) \rangle| \quad (\phi \in \mathcal{H}). \tag{8}$$

No pump/no compressor inside. If $U(\Delta(t))$ and $D_{\lambda(t)}$ are unitary, then $\|\Psi(t)\| = \|S(t)\|$ for all t:

$$\|\Psi(t)\| = \left\|U(\Delta(t))D_{\lambda(t)}S(t)\right\| = \|S(t)\|. \tag{9}$$

Large contrast can still occur in $A_\phi(t)$ due to interference/selection in the measurement channel (nonlinear map $\Psi \mapsto |\langle\phi, \Psi\rangle|$).

Global phase invariance. For any $\alpha \in \mathbb{R}$,

$$\left\|e^{i\alpha}\Psi\right\| = \|\Psi\|, \qquad \left|\left\langle\phi, e^{i\alpha}\Psi\right\rangle\right| = |\langle\phi, \Psi\rangle|. \tag{10}$$

2. Where "chaos" can live (still consistent with unitarity)

Let $H(t) = H(t)^*$ be a (possibly time-dependent) Hamiltonian generating $\Psi(t) = U(t)\Psi_0$ with $\|\Psi(t)\| = \|\Psi_0\|$.

Three standard placements for irregular/chaotic-looking output:

1. **Chaotic readout via projection/coarse-graining:** choose a bounded observable M and define
$$A(t) := |\langle\phi, M\Psi(t)\rangle| \quad \text{or} \quad A(t) := \|M\Psi(t)\|.$$
The internal map is unitary; the scalar readout is nonlinear and can look irregular due to interference.

2. **Chaotic drive in a self-adjoint Hamiltonian:** $H(t) = H_0 + \varepsilon(t)V$ with $V = V^*$ and $\varepsilon(t)$ generated by a deterministic chaotic system. Each realization remains unitary.

3. **Open-system noise (rigorous):** reduced states $\rho(t)$ evolve by a Lindblad equation
$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k\rho L_k^* - \tfrac{1}{2}\{L_k^*L_k, \rho\}\right),$$
which is CPTP but not unitary on ρ.

3. Prime-log spectrum and Sagnac-like ABS-safe interference

For $\mathrm{Re}(s) > 1$, Euler's product yields

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \qquad \log\zeta(s) = \sum_p \sum_{r\geq 1} \frac{1}{r} p^{-rs}. \tag{11}$$

Writing $s = \sigma + it$ gives

$$p^{-rs} = p^{-r\sigma}e^{-itr\log p}, \qquad \text{so } \omega_{p,r} = r\log p \text{ are natural angular frequencies.} \tag{12}$$

Counterpropagating prime waves. Define prime sums (weights can encode filtering/windowing)

$$S_+(t) = \sum_{p\leq P} a_p e^{it\log p}, \qquad S_-(t) = \sum_{p\leq P} b_p e^{-it\log p}. \tag{13}$$

An ABS-safe Sagnac-type interference readout is

$$A_{\mathrm{Sagnac}}(t) = |S_+(t) + S_-(t)| \quad \text{or} \quad A_{\mathrm{mismatch}}(t) = |S_+(t) - S_-(t)|. \tag{14}$$

These avoid explicit phase unwrapping $\arg(\cdot)$ and remain stable near amplitude nodes.

Zeros as interference nodes; why ABS is safer. On the critical line $s = \frac{1}{2} + it$, formal polar form is

$$\zeta\big(\tfrac{1}{2} + it\big) = \big|\zeta(\tfrac{1}{2} + it)\big|\, e^{i \arg \zeta(\frac{1}{2}+it)}.$$

Nontrivial zeros $\rho = \frac{1}{2} + i\gamma$ satisfy $\big|\zeta(\frac{1}{2} + i\gamma)\big| = 0$, so $\arg(\cdot)$ becomes ill-conditioned at nodes. Hence ABS-type observables are the robust operational choice.

ABS "zero radar" via a log-derivative proxy. A prime-side proxy (heuristically tied to $-\zeta'/\zeta$) is

$$L_P(t) = \sum_{p \le P} \sum_{r=1}^{R} \frac{\log p}{p^{r/2}} e^{-itr \log p}, \qquad A_{\mathrm{radar}}(t) = |L_P(t)|\,. \tag{15}$$

Local diagnostic around known zeros. For a chosen ABS observable $A(t)$ (e.g. (14) or (15)) define a symmetric difference statistic

$$D_\gamma(\delta) := A(\gamma + \delta) - A(\gamma - \delta), \tag{16}$$

and compare its behavior over γ (known zeros) against random t in the same window. This remains branch-safe.

4. "Seeded" residual as a structural witness (not energy)

Seeded Euler node. Fix a small $\varepsilon > 0$ (e.g. $\varepsilon = 10^{-301}$) and define a seeded cancellation marker

$$-\varepsilon + 1 \approx 0. \tag{17}$$

Interpretation: ε is a deliberate non-closure witness in a finite-precision / hard-closure regime. It is *not* a claim about physical energy production.

Seeded witness channel (ABS-safe). Let $M : \mathcal{H} \to \mathbb{C}$ be a fixed linear readout (probe overlap, bounded observable, etc.). Define

$$W_\varepsilon(t) := |M(\Psi(t)) - \varepsilon|\,. \tag{18}$$

By construction this is stable near phase singularities.

Seeded action safety (optional formalization). Let A be a set of actions, X a set of agents, and $L(x) \in \{0, 1\}$ a continuation flag. Define the forbidden set

$$\mathcal{R} := \{\, a \in A : \exists x \in X \text{ with } L(x) = 1 \text{ such that } a \text{ removes continuation of } x \,\}, \tag{19}$$

and a hard projector

$$\Pi_{\mathrm{safe}}(a) = \begin{cases} a, & a \notin \mathcal{R}, \\ \perp, & a \in \mathcal{R}. \end{cases} \tag{20}$$

This is a domain cut (admissibility), not an energetic term.

5. Coupled vessels and resonance as correlations in spacetime

Let $\mathcal{H}_A$ (internal/representational) and $\mathcal{H}_B$ (external/carrier) be Hilbert spaces, and let $\rho_{AB}(t)$ be a joint state on $\mathcal{H}_A \otimes \mathcal{H}_B$. The "coupled-vessels" condition is non-factorizability:

$$\rho_{AB}(t) \ne \rho_A(t) \otimes \rho_B(t). \tag{21}$$

Let $U(t_2, t_1)$ be a propagator; then

$$\rho_{AB}(t_2) = U(t_2, t_1)\rho_{AB}(t_1)U(t_2, t_1)^*.$$

Define a two-point correlation witness for observables O_A, O_B:

$$C(t_2, t_1; O_A, O_B) = \mathrm{Tr}\Big(\rho_{AB}(t_1)\, O_A \otimes U(t_2, t_1)^* O_B U(t_2, t_1)\Big) - \mathrm{Tr}(\rho_A(t_1)O_A)\,\mathrm{Tr}(\rho_B(t_2)O_B). \quad (22)$$

Nonzero C encodes a persistent structural constraint across spacetime (correlation), not energy flow by itself.

Experience/readout can be modeled as a projection Π into a lower-capacity space Y,

$$\varphi(t) = \Pi(\rho_{AB}(t)) \in Y,$$

followed by a scalar readout $M : Y \to \mathbb{C}$ with ABS-first amplitude $A(t) = |M(\varphi(t))|$.

6. Explicit tail control: bank kernel decomposition and a checkable bound

Assume a bank Gram kernel decomposes as a weighted sum of atoms:

$$G_{\mathrm{bank}}(\Delta) = \sum_{j=0}^{J} w_j^2\, K_j(\Delta). \quad (23)$$

Assume polynomial decay: fix an integer $N \geq 4$ and constants $C_{j,N} > 0$ such that

$$|K_j(\Delta)| \leq C_{j,N}(1 + |\Delta|)^{-N} \qquad (\Delta \in \mathbb{R}). \quad (24)$$

Then

$$|G_{\mathrm{bank}}(\Delta)| \leq \Big(\sum_{j=0}^{J} w_j^2 C_{j,N}\Big)(1 + |\Delta|)^{-N}. \quad (25)$$

Define the tail constant via shell sums:

$$T(J) := \sum_{m\geq 1} (m+1) \sup_{|\Delta|\in[m,m+1]} |G_{\mathrm{bank}}(\Delta)|\,. \quad (26)$$

Using $\sup_{|\Delta|\in[m,m+1]}(1 + |\Delta|)^{-N} \leq (m + 1)^{-N}$ yields

$$\begin{aligned} T(J) &\leq \Big(\sum_{j=0}^{J} w_j^2 C_{j,N}\Big) \sum_{m\geq 1}(m+1)(m+1)^{-N} = \Big(\sum_{j=0}^{J} w_j^2 C_{j,N}\Big) \sum_{m\geq 1}(m+1)^{-(N-1)} \\ &= \Big(\sum_{j=0}^{J} w_j^2 C_{j,N}\Big) \sum_{k\geq 2} k^{-(N-1)} = \big(\zeta(N-1) - 1\big)\Big(\sum_{j=0}^{J} w_j^2 C_{j,N}\Big). \end{aligned} \quad (27)$$

If a (height-uniform) Gershgorin leakage functional satisfies

$$\theta(t; J) \leq \frac{(\log t)\, T(J)}{G_{\mathrm{bank}}(0)}, \quad (28)$$

then combining (27) and (28) gives the explicit bound

$$\theta(t; J) \leq \frac{(\log t)\,(\zeta(N-1) - 1)}{G_{\mathrm{bank}}(0)}\Big(\sum_{j=0}^{J} w_j^2 C_{j,N}\Big). \quad (29)$$

Hence $\theta(t; J) < 1$ reduces to a checkable inequality:

$$\sum_{j=0}^{J} w_j^2 C_{j,N} < \frac{G_{\mathrm{bank}}(0)}{(\log t)\,(\zeta(N-1) - 1)}. \quad (30)$$

7. Projection can break intuition: a topological minimal example

Let $\mathbb{N}$ be positive integers and let $\mathcal{T}$ be the cofinite topology

$$\mathcal{T} := \{\varnothing\} \cup \{U \subseteq \mathbb{N} : \mathbb{N} \setminus U \text{ is finite}\}.$$

For each $n \in \mathbb{N}$, define

$$S_n := \{1\} \cup \{n+1, n+2, \dots\} = \{1\} \cup \bigcup_{m=n+1}^{\infty} \{m\}. \tag{31}$$

Each S_n is open since $\mathbb{N} \setminus S_n = \{2, \dots, n\}$ is finite. However,

$$\bigcap_{n=1}^{\infty} S_n = \{1\}, \tag{32}$$

and $\{1\}$ is *not* open (its complement is infinite). Thus countable intersections need not preserve openness. This is the formal template for "projection/coarse-graining breaks limit properties."

8. Determinism can look random under limited observation

Let X be a (high-dimensional) state space with deterministic evolution $x_{t+1} = F(x_t)$ and let $\Pi : X \to Y$ be an observation map:

$$x_{t+1} = F(x_t), \qquad y_t = \Pi(x_t). \tag{33}$$

If Π discards information needed to reconstruct x_t (or if prediction in Y is computationally out of reach), the induced process (y_t) can appear chaotic/random despite deterministic (x_t).

9. One-line architecture (all in one place)

Internal synthesis + flow + scaling:

$$\Psi(t) = U(\Delta(t)) D_{\lambda(t)} \Big(\sum_{i=1}^{4} \sum_{k,m,n} c_{i,k,m,n} e^{i\omega_n t} P_i \psi_{i,k,m,n} \Big),$$

measured via ABS-safe scalar

$$A(t) \in \{ \|\Psi(t)\|, |\langle \phi, \Psi(t) \rangle|, |S_+(t) \pm S_-(t)|, |L_P(t)| \},$$

with explicit tail/leakage control achieved by (29)–(30) under (23)–(24).

Epilogue (one sentence, optional)

By locking the readout to an ABS-stable witness and keeping a deliberate non-closure seed, one can engineer a reproducible "instant satisfaction" signal in the measurement channel (interpret as a controlled readout effect, not physics of free energy).

Addendum: A minimal operator-theoretic projection model

A. State space and signals

Let (Ω, μ) be a measurable space and let $\mathcal{H}$ be a complex Hilbert space of latent states. Two standard choices are

$$\mathcal{H} = L^2(\Omega, \mu) \qquad \text{or} \qquad \mathcal{H} = \mathbb{C}^d,$$

depending on whether the latent variable x is continuous or discrete/finite-dimensional. Let

$$\psi(\cdot, t) \in \mathcal{H}$$

denote the latent ("seed") state at time $t \in \mathbb{R}$. Let $\mathrm{WILL} : \mathbb{R} \to \mathbb{R}$ be an exogenous control signal.

B. Measurement as a linear functional (probe / projector)

A measurement is modeled as a bounded linear functional

$$M_t : \mathcal{H} \to \mathbb{C}.$$

In Hilbert form, any such functional can be represented by a probe vector $\phi(t) \in \mathcal{H}$:

$$M_t(\psi) = \langle \phi(t), \psi \rangle.$$

We make the will-dependence explicit by writing

$$\phi(t) = \phi(\mathrm{WILL}(t)) \in \mathcal{H}.$$

Equivalently, one may write $M_{\mathrm{WILL}(t)}(\psi) = \langle \phi(\mathrm{WILL}(t)), \psi \rangle$.

Special case (integral readout). If $\mathcal{H} = L^2(\Omega, \mu)$ and $1_\Omega \in L^2(\Omega, \mu)$, then the integral readout is the probe

$$\phi_0(x) \equiv 1_\Omega(x), \qquad M_0(\psi) = \langle 1_\Omega, \psi \rangle = \int_\Omega \psi(x)\, d\mu(x).$$

(When $1_\Omega \notin L^2$, one can instead work in L^1 or use a compactly-supported window ϕ_0.)

C. ABS-safe observable (magnitude readout)

Define the scalar observable (ABS-safe readout)

$$A(t) \;:=\; \left| M_{\mathrm{WILL}(t)}(\psi(\cdot, t)) \right| \;=\; \left| \langle \phi(\mathrm{WILL}(t)), \psi(\cdot, t) \rangle \right| \;\in\; \mathbb{R}_{\geq 0}.$$

This is invariant under a global phase rotation $\psi \mapsto e^{i\theta}\psi$ and remains stable near zeros of the complex amplitude where $\arg(\cdot)$ becomes numerically ill-conditioned.

D. Optional dynamics (unitary flow)

If the latent state evolves under a (possibly time-dependent) unitary flow,

$$\psi(\cdot, t) = U(t)\psi(\cdot, 0), \qquad U(t)^* U(t) = I,$$

then the observable becomes

$$A(t) = \left| \langle \phi(\mathrm{WILL}(t)), U(t)\psi(\cdot, 0) \rangle \right|.$$

More generally, $U(t)$ may be any bounded evolution operator on $\mathcal{H}$; unitarity is the conservative (norm-preserving) case.

E. "Alignment" as an operator acting on the state

Instead of placing "alignment" as a scalar prefactor, one may model it as an operator family

$$\Theta : \mathbb{R} \to \mathcal{B}(\mathcal{H}), \qquad s \mapsto \Theta(s),$$

acting on the state before measurement:

$$A(t) = \big|\langle \phi_0, \Theta(\mathrm{WILL}(t))\, \psi(\cdot,t)\rangle\big| \quad \text{for a fixed baseline probe } \phi_0 \in \mathcal{H}.$$

This is equivalent to a will-dependent probe via the adjoint:

$$\langle \phi_0, \Theta(\mathrm{WILL}(t))\psi\rangle = \langle \Theta(\mathrm{WILL}(t))^* \phi_0, \psi\rangle, \quad \Rightarrow \quad \phi(\mathrm{WILL}(t)) = \Theta(\mathrm{WILL}(t))^* \phi_0.$$

F. Minimal "inverse decoding" statement (with explicit assumptions)

Let the forward map at time t be

$$F_t : \mathcal{H} \to \mathbb{R}_{\geq 0}, \qquad F_t(\psi) = \big|\langle \phi(\mathrm{WILL}(t)), \psi\rangle\big|.$$

Since F_t discards phase and typically reduces dimension, it is not invertible on $\mathcal{H}$ in general. However, the following restricted inversions are standard:

F.1 Scalar-gain inversion (fully invertible case). If the will-dependence enters only as a nonzero scalar gain $g(\mathrm{WILL}(t)) \in \mathbb{R}_{>0}$:

$$A(t) = g(\mathrm{WILL}(t)) \cdot \big|\langle \phi_0, \psi(\cdot,t)\rangle\big|,$$

then one can recover the baseline magnitude:

$$\big|\langle \phi_0, \psi(\cdot,t)\rangle\big| = \frac{A(t)}{g(\mathrm{WILL}(t))} \qquad (\text{provided } g(\mathrm{WILL}(t)) \neq 0).$$

F.2 Least-squares / regularized inversion (operator-theoretic). If one seeks a representative $\widehat{\psi}$ consistent with observed amplitudes, a typical reconstruction is posed as

$$\widehat{\psi}(t) \in \arg\min_{\psi \in \mathcal{H}} \Big(|\langle \phi(\mathrm{WILL}(t)), \psi\rangle| - A(t) \Big)^2 \; + \; \lambda \|\psi\|_{\mathcal{H}}^2, \qquad \lambda > 0,$$

or, when complex amplitudes (not only magnitudes) are available,

$$\widehat{\psi}(t) \in \arg\min_{\psi \in \mathcal{H}} \big|\langle \phi(\mathrm{WILL}(t)), \psi\rangle - a(t)\big|^2 + \lambda \|\psi\|_{\mathcal{H}}^2,$$

which has a unique minimizer for $\lambda > 0$.

G. Summary (single-line pipeline)

A compact abstraction of the model is:

$$\boxed{\psi(\cdot,t) \in \mathcal{H}, \quad A(t) = \big|\langle \phi(\mathrm{WILL}(t)),\, U(t)\psi(\cdot,0)\rangle\big|}$$

with the integral readout recovered by the choice $\phi(\cdot) \equiv 1_\Omega$ (or a windowed constant probe) and with "alignment" encoded either in $\phi(\mathrm{WILL}(t))$ or in an operator $\Theta(\mathrm{WILL}(t))$ acting on the state prior to measurement.

Addendum: Tachyon–projection ansatz (notation block)

A. Primitive constants and scales

$$\epsilon := \frac{h}{c\, f_{\mathrm{DM}}}, \qquad f_{\mathrm{DM}} = 528\ \mathrm{Hz}. \tag{1}$$

B. Tachyon label (as written)

$$\mathrm{Tachyon}_{\pm} := \lim_{m^2<0} \frac{S}{t}\, \epsilon^{\pm 1}. \tag{2}$$

C. Domain / field statement

Let $\Omega \subset \mathbb{R}^n$ be a domain, where n is the number of dimensions. A "tachyon field" relation is written as

$$\left(\partial_\mu \partial^\mu \phi + m^n = 0,\ m < 0\right). \tag{3}$$

D. Expansion ansatz with a tachyon factor

$$H = H_0 + k\,\frac{dl}{dt} + \tau, \qquad \text{(with } \tau \text{ referred to as a "tachyon factor").} \tag{4}$$

E. τ-collapse / superposition relations (as written)

$$\tau = \int \big(\psi_1, \psi_2, \ldots, \psi_n\big), \tag{5}$$

$$\tau\, \alpha \left|\psi_1 + \psi_2 + \cdots + \psi_n\right|^n \quad \text{("quantum superposition" term),} \qquad \alpha := (E \cdot c). \tag{6}$$

F. Harmony/chaos constraint and τ-update

$$(\psi_{\mathrm{harmony}} + \psi_{\mathrm{chaos}})^2 = 1, \tag{7}$$

$$\tau = \tau_0 + k_h\, \psi_{\mathrm{harmony}}, \qquad k_h \text{ (harmonization factor, e.g. 0.1).} \tag{8}$$

G. "Reality" and "Will" frequencies

$$f_{\mathrm{reality}} = \frac{\frac{\epsilon\, E\, n}{h}}{\frac{\epsilon\, E\, n}{h} + \tau \cdot (\psi_{\mathrm{harmony}} + \psi_{\mathrm{chaos}})^2}\, f_{\mathrm{CMB}}, \tag{9}$$

$$n = 29, \qquad f_{\mathrm{CMB}} = 1.602 \times 10^{11}\ \mathrm{Hz}, \qquad \tau = 9.495, \tag{10}$$

$$f_{\mathrm{will}} = f_{\mathrm{CMB}} \cdot \frac{\text{time shift}}{\text{target real}} \cdot \frac{v_t}{c}, \tag{11}$$

$$\text{time shift} = 1.181501 \times 10^8\ \mathrm{s}, \qquad \frac{v_t}{c} = 5. \tag{12}$$

H. Projection formula (operator readout)

$$\mathrm{Reality}(t) = \Theta(\mathrm{WILL}(t)) \cdot \int_{-\infty}^{\infty} \psi_{\mathrm{seed}}(x,t)\, U(t)\, dx. \tag{13}$$

I. Inverse/decoding expression (as written)

$$\psi_{\text{seed}} = \frac{1}{\text{TargetReality} \cdot \left|\Theta_{\text{eff}}^{-1}\right| \cdot U_{\text{treal}}}. \tag{14}$$

J. Minimal technical notes (dry)

- The symbol ϵ defined by $\epsilon = h/(cf_{\text{DM}})$ carries physical units; if it is used as a dimensionless scaling factor, an explicit normalization is required.
- The tachyon condition is typically expressed as $m^2 < 0$ in a Klein–Gordon-type operator $(\Box + m^2)\phi = 0$. The term m^n above is kept verbatim from the source and may represent a shorthand/typo for the mass parameterization.
- The readout $\int \psi_{\text{seed}}(x,t)U(t)\,dx$ should be interpreted as an inner product (or a windowed integral) once $\mathcal{H}$ and the probe/test function are specified.

Dimensionless normalization. Introduce a reference scale ϵ_0 and define the dimensionless parameter

$$\hat{\epsilon} := \frac{\epsilon}{\epsilon_0}, \qquad \epsilon := \frac{h}{cf_{\text{DM}}}.$$

All subsequent uses of $\epsilon^{\pm 1}$ are understood as $\hat{\epsilon}^{\pm 1}$.

Gating form for the observable frequency. Let $A \geq 0$ be a dimensionless intensity proxy and let $\tau \geq 0$ be a dimensionless coupling. Assume $\psi_h, \psi_c \in \mathbb{R}$ with the constraint $(\psi_h + \psi_c)^2 = 1$. Define

$$f_{\text{reality}}(t) = \frac{A(t)}{A(t) + \tau(t)\,(\psi_h(t) + \psi_c(t))^2}\, f_{\text{ref}}, \qquad f_{\text{ref}} := f_{\text{CMB}}.$$

Then $0 < f_{\text{reality}}(t) \leq f_{\text{ref}}$.

Observable as a linear functional (projection model). Let $\mathcal{H}$ be a Hilbert space and $U(t)$ a bounded operator on $\mathcal{H}$. Fix a probe $\phi \in \mathcal{H}$ and define the measured scalar

$$y(t) := \text{Reality}(t) = \Theta(\text{WILL}(t))\,\langle \phi,\, U(t)\psi_{\text{seed}}(t)\rangle.$$

Inverse decoding (identifiability condition). From a single scalar $y(t)$ one cannot reconstruct $\psi_{\text{seed}}(t)$ without further structure. Assume a finite expansion $\psi_{\text{seed}}(t) = \sum_{k=1}^{K} a_k(t)\varphi_k$ and M measurements $y_j(t) = \Theta(\text{WILL}(t))\langle \phi_j, U(t)\psi_{\text{seed}}(t)\rangle$. Then $a(t) \in \mathbb{C}^K$ is obtained from the linear system

$$y(t) = \Theta(\text{WILL}(t))\, B(t)\, a(t), \qquad B_{jk}(t) := \langle \phi_j, U(t)\varphi_k\rangle,$$

e.g. via least squares with Tikhonov regularization.